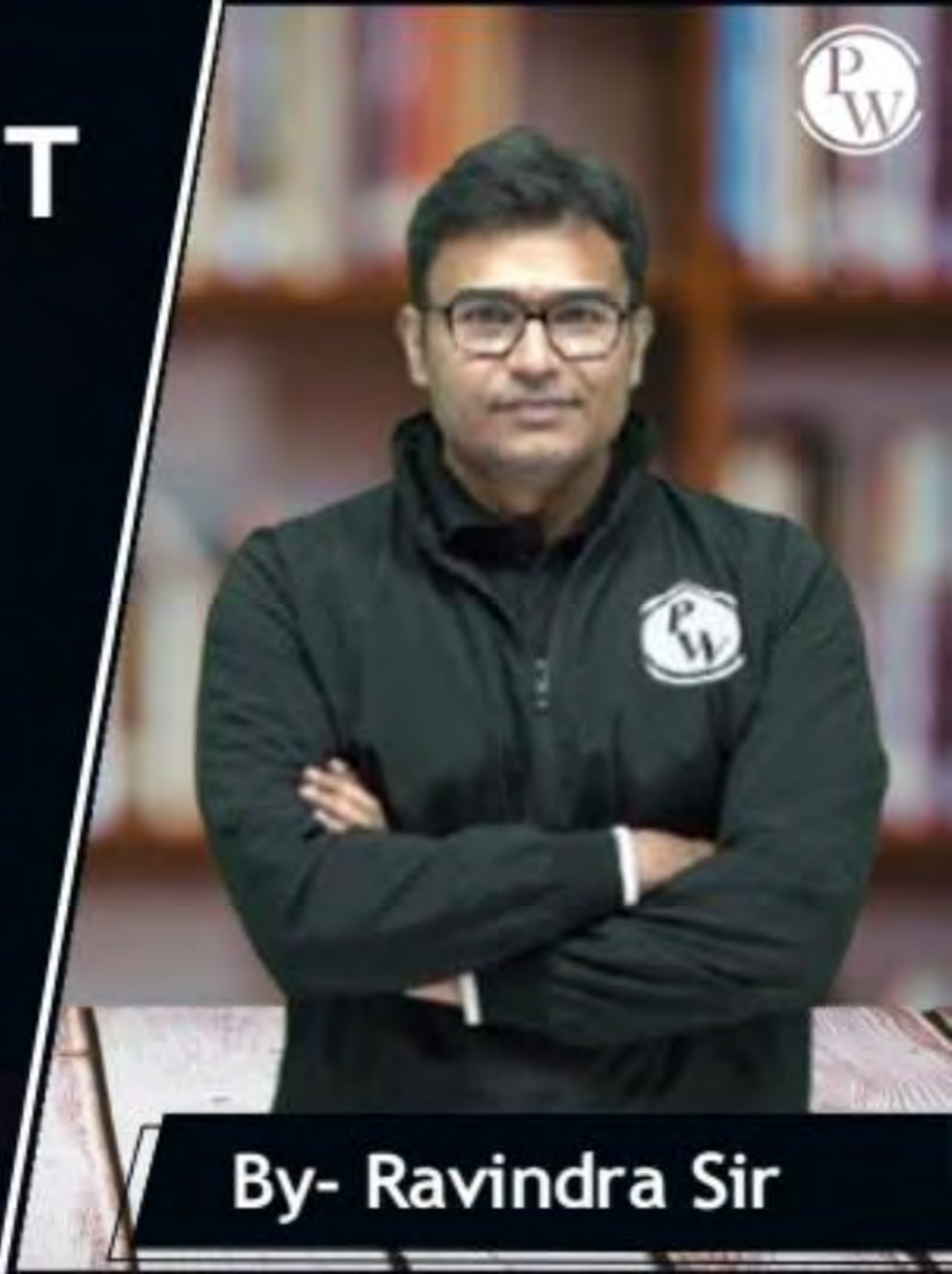
Computer Science & IT

COMPILER DESIGN
(CD)

RunTime Environments

One Shot



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

SDT

Topics to be Covered



- Topic
- Topic
- Topic

RTE

Join Telegram

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Inspiring Stories : Mula Gabharu

Mula Gabharu – The Queen Who Raised Women Soldiers

Background: Mula Gabharu was the daughter of an Ahom king in Assam in the early 1500s.

Struggles: Her husband fell fighting an invading army. Her people were helpless and losing hope.

Achievements: She trained women warriors, led them into battle with courage, and confronted the enemy herself before being martyred.

Impact: If she could turn grief into action and lead women into battle, you can turn doubt into study and lead your rank up.



Inspiring Stories : B. N. Jagatap

B. N. Jagatap

Background: Grew up in a modest family in rural Maharashtra with no strong science background at home.

Education: PhD in Nuclear Magnetic Resonance (NMR) from IISc Bangalore.

Career Achievements: Improved NMR methods to study molecules in great detail, helping drug design and materials research.

Impact: Made high-end science tools more effective for Indian research labs and universities.



Inspiring Stories : Veeramangai Kuyili

Veeramangai Kuyili – The Woman Who Became a Human Torch

Background: Kuyili was a trusted soldier of queen Velu Nachiyar from Tamil Nadu in the 1700s.

Struggles: She worked without praise and lived in a time when women were rarely seen in battle.

Achievements: During one attack, she covered her body in ghee, set herself on fire, and jumped into the British armory. Her act became a key turning point in the war.

Impact: If she could light herself on fire to save her queen, you can light up that answer sheet with focus



Inspiring Stories : K. J. Rao

K. J. Rao

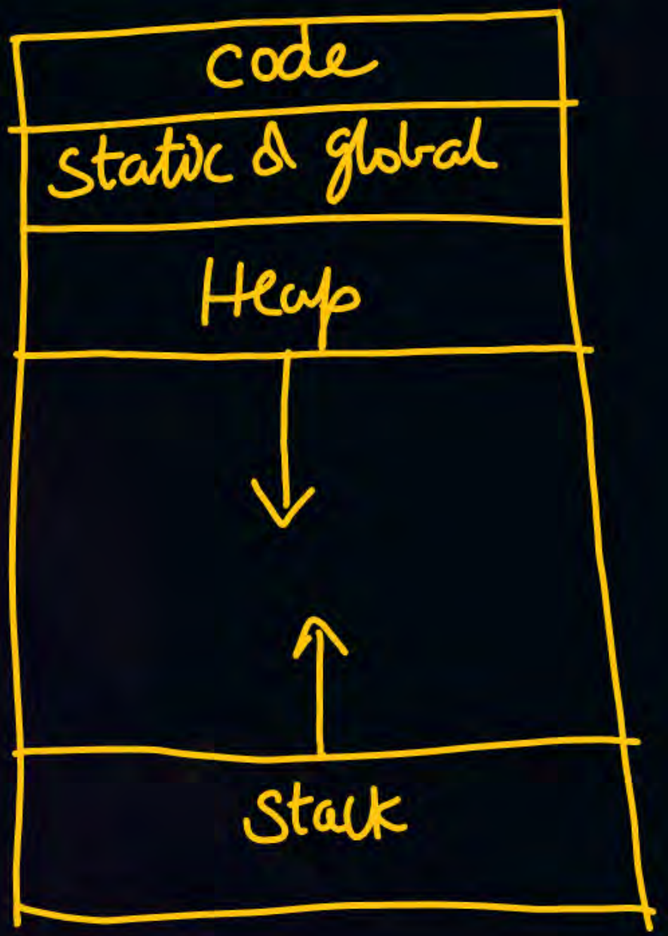
Background: Born in a small town in Karnataka with very limited means. Had to work hard to afford studies.

Education: PhD in Solid State Chemistry from IISc Bangalore.

Career Achievements: Developed new ceramic materials that can store energy and work in extreme heat.

Impact: Helped India make better energy storage systems and heat-resistant tools for industry.

Process:



Storage allocation Strategies:-

1) Static:-
   a) allocation is done at Compile time.
   b) Bindings do not change at Runtime
   c) one activation record per procedure.

Dis:- a) Recursion is not Supported
   b) Size of the data objects must be known at compile time.
   c) Data structures cannot be created dynamically

2) Stack :-

Whenever a new activation begins, activation record is pushed on to the Stack. Whenever activation ends, activation record is popped.

Local variables are bound to fresh storage

Dis :- local variables cannot be retained once the activation ends

3) Heap :-

Allocation and deallocation can be done in any order

Dis:- Heap management is a overhead

Summary :-

Activations can have

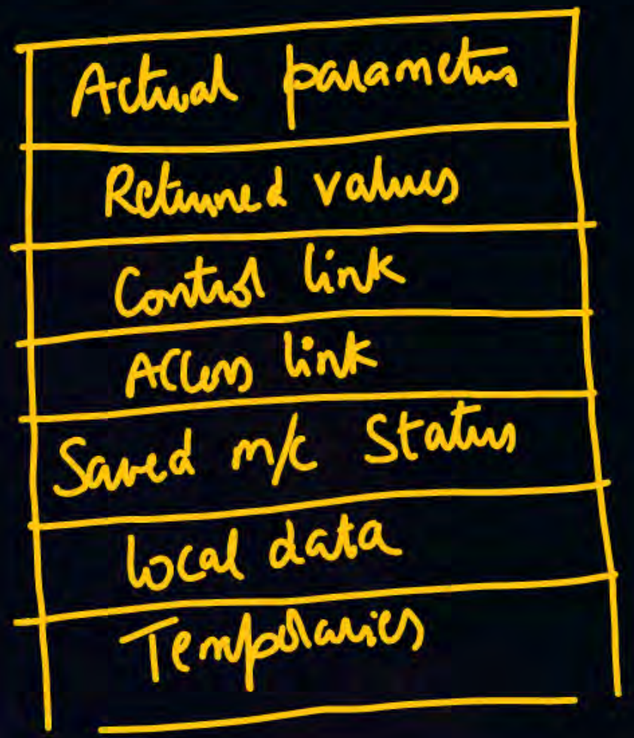
i) Permanent lifetime in case of static allocation

iii) Nested lifetime in case of stack allocation



(iii) Arbitrary lifetime in case of heap allocation

Activation Record / Stack frame



Temporaries:- Temporary variables arise from evaluation of expressions

In case there is no space in registers, they will be stored in activation record.

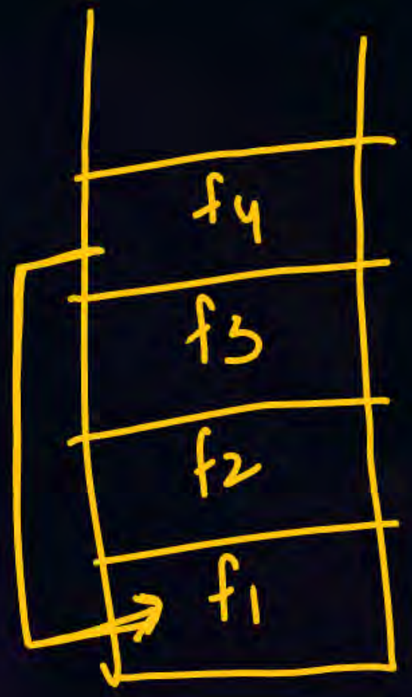
Local data - It belongs to the procedure whose activation record it is.

Saved m/c States: It contains the information about the state of the machine just before the call to the procedure. This information typically includes the return address (value of the program counter, to which procedure must return) and the contents of the registers that were used by the calling procedure and must be restored when the return occurs.

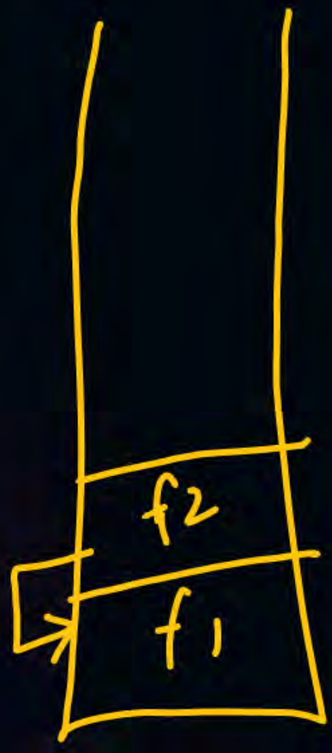


Access link

An access link may be needed to locate data needed by the called function but found elsewhere, e.g. in another activation record



Control link :- It points to the activation record of the caller



Returned value:

Space for the return value of the called function, if any. Not all called procedures returns a value, and if one does we prefer to place that value in a register for efficiency.

Actual parameters:-

The actual parameters are passed by the calling procedure
Commonly, these values are not placed in the activation record
but rather in registers; when possible for greater efficiency. However
we show a space in activation record if registers are full

THANK - YOU